

HW2

1 Question 1

1.1 Part a

Load the `hw2data.csv` data and create a scatterplot of `x` and `y`.

1.2 Part b

Fit an OLS linear regression model predicting `y` given `x`. Interpret the regression coefficient of `x`.

1.3 Part c

Recreate a scatterplot of `x` and `y`, this time specify the color of the points using the `subject`.

1.4 Part d

Fit a random intercept model for `y` given `x`. Interpret the regression coefficient for the fixed effect of `x`.

1.5 Part e

Fit a mixed effects model for `y` given `x`, with a random intercept and a random slope. Interpret the regression coefficient for the fixed effect of `x`.

2 Question 2

2.1 Part a

Assess whether the conditional residuals are consistent with the conditions for use for the mixed effects model fit in Question 2(e).

2.2 Part b

Assess whether the marginal residuals are consistent with the conditions for use for the mixed effects model fit in Question 2(e).

2.3 Part c

Assess whether the random effects are consistent with the conditions for use for the mixed effects model fit in Question 2(e).

2.4 Part d

Compare the models fit in for the mixed effects model fit in Question 2 (d) and Question 2 (e). Does adding the random slopes discernibly improve model fit?

3 Question 3

Dicken et al. (2025) investigate whether eating ultraprocessed foods affects weight loss even when you are following official healthy eating guidelines. The researchers assigned participants eight-week diets that are consistent with the National Health Service Eatwell Guide that outlines government policy about balanced diets for the United Kingdom:

- Minimally Processed Foods: Fresh vegetables, grains, and unprocessed meats
- Ultraprocessed Foods: Packaged snacks, cereals, and prepared meals

The researchers assigned participants to one of two randomization arms:

- **Randomization Arm 1:**
 - Week 1 - Week 2: Baseline Measurements
 - Week 3 - Week 10: Ultraprocessed Foods Diet (provided all meals)
 - Week 11 - Week 14: No Specified Diet
 - Week 15 - Week 16: Baseline Measurements (provided all meals)
 - Week 17 - Week 24: Minimally Processed Foods
- **Randomization Arm 2:**
 - Week 1 - Week 2: Baseline Measurements
 - Week 3 - Week 10: Minimally Processed Foods (provided all meals)
 - Week 11 - Week 14: No Specified Diet
 - Week 15 - Week 16: Baseline Measurements
 - Week 17 - Week 24: Ultraprocessed Foods Diet (provided all meals)

Their work aims to evaluate whether it is just about the nutrients – what is in the food – or if it is the nutrients *and* how they are prepared. Specifically, they aimed to model the percent change in weight based on the diet followed, the randomization arm, and whether the participants worked the night shift. The researchers note that a diet may be less effective if it is the second diet they are on, and they want to assess that in the model.

3.1 Part a

Write out the observation-level regression model that the researchers should use using mathematical notation. Define variables you use in the equation in an itemized list under the regression equation.

3.2 Part b

Write out the subject-level regression model that the researchers should use using mathematical notation. Define variables you use in the equation in an itemized list under the regression equation.

3.3 Part c

How can the researchers assess whether either diet is discernibly better? Write out an analysis plan for what specifically you would do once the data are collected.

3.4 Part d

How can the researchers assess whether a diet may be discernibly less effective if it is the second diet? Write out an analysis plan for what specifically you would do once the data are collected.

3.5 Part e

How can the researchers assess whether there was a discernible difference across randomization arms? Write out an analysis plan for what specifically you would do once the data are collected.

References

Dicken, Samuel J, Friedrich C Jassil, Adrian Brown, Monika Kalis, Chloe Stanley, Chaniqua Ranson, Tapiwa Ruwona, et al. 2025. “Ultraprocessed or Minimally Processed Diets Following Healthy Dietary Guidelines on Weight and Cardiometabolic Health: A Randomized, Crossover Trial.” *Nature Medicine* 31 (10): 3297–3308.